RISC-V LOCAL INFERENCE

Qwen3.5 Inference on the SpaceMiT K1

Infrastructure Optimization and Performance Evaluation

Evidence cutoff: October 16, 2026 (Asia/Singapore)

Platform	MUSE-Pi-Pro, K1/X60, nominal 16 GB RAM
Models	Qwen3.5 2B and 4B
Implementation	SpaceMiT llama.cpp fork
Package	Technical report and evidence appendix

This report compares one compiled default build with the original compatibility baseline. Long-input results have one pair per model. Formal acceptance gaps remain.

github.com/MoyaMryia/riscv-accl

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Abstract

The final build completes one request 1.24 to 3.59 times faster on 2B and 1.41 to 4.72 times faster on 4B. The comparison is against the original vendor commit `5ad05d8` with two compatibility changes. The gain increases with input length. Server memory stays the same: the two arms differ by less than 3 MiB.

Model	Input tokens	Pairs	Request time, s	Faster by	TTFT reduction	Decode, tokens/s
2B	512	3	43.031 / 34.714	1.24x	17.96%	3.387 / 4.292
2B	2048	3	136.554 / 97.169	1.41x	29.41%	3.108 / 4.186
2B	8192	1	875.359 / 426.221	2.05x	51.77%	2.239 / 3.600
2B	16384	2	2868.832 / 1057.226	2.71x	63.43%	1.659 / 2.914
2B	32768	1	10822.886 / 3014.282	3.59x	72.27%	1.051 / 2.141
4B	512	3	113.128 / 80.346	1.41x	21.72%	1.315 / 2.133
4B	2048	3	411.793 / 246.579	1.67x	39.77%	1.133 / 1.965
4B	8192	1	3323.894 / 1170.324	2.84x	65.17%	0.724 / 1.485
4B	16384	2	11388.985 / 3033.905	3.75x	73.56%	0.490 / 1.117
4B	32768	1	42294.220 / 8952.505	4.72x	78.93%	0.291 / 0.752

Each slash separates original and final means. "Faster by" is the original request time divided by the final request time. Both arms use identical Q4_0 weights, F16 cache, inputs, and 65 output tokens. Sections 5 and 5.1 give the complete tables.

Version `k1-final-20261013-r3` is one build for both models. It sets the selected parameters in the source. Normal startup requires no performance flags or experiment variables. The build passes native checks, state checks, natural code answers, and relocation checks. The default is the balanced choice of a frozen search. The Scope section at the end lists the sample counts and the open acceptance items.

1. Requirements and status

Requirements.md is the sole current acceptance reference. It requires both specified model bases, local inference, the vendor fork, matched improvements, and reproducible deployment materials. It does not set an 8 GB device requirement or a 32k/64k context requirement. The gap report retains the complete requirement map.

Status	Verified scope or limit
Implemented	One complete source tree, one production build, fixed defaults, dependency checks, and a common launcher
Verified	Both current GGUF files run locally on K1 without a cloud API
Verified	28 ordinary-default performance requests, 12 capacity-extension requests, and six complete natural code answers
Verified	Exact native dumps, 64 short state records, and two relocated model launches
Pending	Full official model origin and reproducible conversion of both final weights
Pending	Repeated long-context stability, long state acceptance, and the fixed multi-turn quality suite
Not implemented in this release	GPU acceleration, KV compression, and memory reduction through page allocation

No external competition submission is recorded.

2. Platform and software identity

The tested board is a MUSE-Pi-Pro with eight K1/X60 CPU cores. Its nominal RAM is 16 GB; Linux reports approximately 15.528 GiB. Recorded CPU frequency is 1.6 GHz. Tests use an ordinary user in tmux under the common board lock. No device permission or system change forms part of this campaign.

Item	Identity
Final version	`k1-final-20261013-r3`
Vendor source	`a990751d55a4c54acf2bb77d44282c2093652359`
Complete source count	3271 files
Final patch SHA-256	`b03d193f618d8303ce563c4fcd4e03726a1f6f45 2dcc10ecedb1a7938550477b`
Server SHA-256	`f254f174919c0a97ef34366f116705f6d22574ca b4c2bfc39928c0d315b5bfbd`
Policy SHA-256	`aeef14e2356dc9e33b1fc6ab595545eb67af8189 4f2c7cbd282948628453bdb0`
Compiler	GCC 14.2.0 with vendor IME support
Runtime	SPERT 0.6.2 and `libspine_tcm.so.3.0.1`; exact file hashes remain in the build receipt
Build	Release, IME1 enabled, OpenMP disabled, Native disabled, ZBA enabled, repack disabled, CPU backend

The two model files use Q4_0 weights and Q4_0 embeddings. Their names contain MTP tensors, but this release uses direct decode. The package does not contain weights or vendor runtime libraries.

Model	GGUF SHA-256
`Qwen3.5-2B-MTP-Q4_0-embQ4_0-dv64k.gguf`	`56f0ddd90dfa0e2456af6cb4718ece419678d21a 75d65e27353319b5e431a2fa`
`Qwen3.5-4B-MTP-Q4_0-embQ4_0-dv64k.gguf`	`0adf6cce5df53921606033c39a13fae2054543b7 8af0e7ff98860e86f078ac0f`

The default-build TCM trace verifies device access during two short requests. Both servers load the fixed library, but hardware TCM access fails with `EACCES`. No `/dev/tcm` mapping appears in 13 recorded process maps. The shared barrier falls back to heap memory because `/dev/tcm_sync_mem` is absent. The new user API rejects fake TCM; hardware read/write tests remain pending. These traced requests establish no new performance or answer-quality result.

3. Original baseline and compatibility

The original source is `5ad05d8796cb094ee23a3a0152874f8a2b3060f8` from the SpaceMiT fork. Its K1 build includes an IME2 file and fails. The first compatibility change removes that single CMake source-list entry. The resulting build cannot load the current reduced MTP draft head. The second change loads `d2t` and uses the stored draft head width. Neither arm enables the draft decoder.

Only the CMake list and the tensor loader in `src/models/qwen35.cpp` change. The main output head, graph code, and all compute kernels retain their original bytes. The compatibility patch SHA-256 is `2e8c40a4 a056787380e37750bab005777580978d3d9a58d901e911dc12fe01d7`. Both failures and both compatibility records remain in the evidence. This is a compatibility baseline, not an unmodified binary.

Both arms use four threads, batch 32, context 12288, one slot, and F16 K/V. They use the same compiler and runtime libraries. The measured gain spans the original software state to the final software state. It excludes historical weight quantization changes. It does not isolate each individual kernel contribution.

4. Balanced default selection

The frozen search uses one verified RC1 build and 104 requests. Four preflight requests check both models on the original and candidate builds. The mask stage tests eight combinations on two models and two lengths, with two rounds. The parameter stage tests 2/4 threads and batch sizes 16/32/64/128. Four natural answers complete the selection checks. The parameter stage retains four threads and batch 32. No additional parameter confirmation or fallback is required.

The balanced score gives equal weight to both models, both lengths, TTFT, and decode time. TTFT means client time to the first output token. A difference within 3% of the lowest score is unresolved under the frozen rule. This rule prefers fewer enabled candidates within that interval. Mask 7 has the lowest observed score, 0.8904876472483813. Mask 3 scores 0.915536301962804 and lies within that interval. The rule therefore selects mask 3. The selection score uses the same-build C0 control, not the original baseline.

Compiled default	Value
Threads and batch threads	4 / 4
Batch and physical batch	32 / 32
Context and slots	12288 / 1
Cache and attention	F16 K/V, flash attention enabled
GPU layers and decode	0, direct decode
Common paths	Wide RVV attention and IME route 3 enabled
Candidate mask	3: B1-r2 and C1 enabled; C2-r2 disabled

The launcher clears inherited experiment variables. The measured final server receives only the model, alias, host, and port. Runtime logs and server properties confirm the actual defaults.

5. Matched performance results

Performance requests use complete fixed input token lists and 65 fixed output tokens. A single-token bias forces identical decode histories across versions. This synthetic workload measures performance, not answer quality. Natural answers use a separate request without this bias.

All requests have zero prompt-cache reuse. The tests do not clear the operating-system file cache. Each server starts before one request; server load time is outside request timing. All 28 requests complete and generate exactly 65 tokens. Input hashes, output tokens, full text, and finite timings match the protocol.

Short shapes use three pairs with alternating arm order. Long shapes use one pair per model, with opposite order between models. Tables show arithmetic means of per-request metrics. Each slash separates original and final values, in that order. Time reductions and throughput increases use different formulas.

Model	Input	Pairs	TTFT, s	Server prefill, s	Prefill, tokens/s
2B	512	3	23.835 / 19.556	23.831 / 19.551	21.485 / 26.188
2B	2048	3	115.639 / 81.633	115.632 / 81.626	17.712 / 25.090
2B	8192	1	846.321 / 408.162	846.303 / 408.145	9.680 / 20.071
4B	512	3	63.697 / 49.865	63.692 / 49.860	8.039 / 10.269
4B	2048	3	354.441 / 213.497	354.434 / 213.490	5.778 / 9.593
4B	8192	1	3234.172 / 1126.535	3234.155 / 1126.518	2.533 / 7.272

Model	Input	Decode, tokens/s	Throughput change	Request, s	Peak RSS, MiB
2B	512	3.387 / 4.292	+26.69%	43.031 / 34.714	2471.97 / 2469.73
2B	2048	3.108 / 4.186	+34.68%	136.554 / 97.169	2472.29 / 2469.96
2B	8192	2.239 / 3.600	+60.82%	875.359 / 426.221	2473.30 / 2471.19
4B	512	1.315 / 2.133	+62.17%	113.128 / 80.346	5342.20 / 5340.58
4B	2048	1.133 / 1.965	+73.37%	411.793 / 246.579	5342.20 / 5340.88
4B	8192	0.724 / 1.485	+104.91%	3323.894 / 1170.324	5342.88 / 5342.11

Model	Input	TTFT reduction	Prefill throughput change	Decode time reduction
2B	512	17.96%	+21.89%	21.04%
2B	2048	29.41%	+41.66%	25.72%
2B	8192	51.77%	+107.35%	37.82%
4B	512	21.72%	+27.74%	38.34%
4B	2048	39.77%	+66.02%	42.32%
4B	8192	65.17%	+187.09%	51.20%

RSS is resident memory for the server process. The table reports the mean of sampled request peaks, with a 0.2 s sample interval. It does not measure whole-machine memory or an instantaneous hardware peak. The independent summary retains means, medians, minima, maxima, and each request identity. No shape has a clear TTFT or decode-time regression under the frozen threshold. The threshold is the greater of 3% and the full original-arm variation range.

5.1. Final-build capacity extension

A separate frozen protocol extends capacity to 20480 or 36864 for actual 16384 or 32768 input tokens. The final server receives only the capacity flag in addition to model, alias, host, and port. Its optimization defaults remain unchanged; ordinary context remains 12288. Both models complete all requests with 65 matched output tokens and zero prompt reuse. The test uses two pairs per model at 16384 inputs and one pair per model at 32768 inputs.

Model	Input	Pairs	TTFT, s	Prefill, s	Prefill, tokens/s
2B	16384	2	2829.646 / 1034.907	2829.614 / 1034.876	5.797 / 15.832
2B	32768	1	10761.012 / 2983.902	10760.952 / 2983.845	3.045 / 10.982
4B	16384	2	11256.195 / 2975.716	11256.164 / 2975.685	1.456 / 5.506
4B	32768	1	42070.823 / 8866.022	42070.765 / 8865.966	0.779 / 3.696

Model	Input	Decode, tokens/s	Request, s	Peak RSS, MiB
2B	16384	1.659 / 2.914	2868.832 / 1057.226	2571.781 / 2569.617
2B	32768	1.051 / 2.141	10822.886 / 3014.282	2767.641 / 2766.426
4B	16384	0.490 / 1.117	11388.985 / 3033.905	5600.367 / 5600.611
4B	32768	0.291 / 0.752	42294.220 / 8952.505	6117.164 / 6117.809

Model	Input	TTFT reduction	Prefill gain	Decode gain
2B	16384	63.43%	+173.13%	+75.59%
2B	32768	72.27%	+260.64%	+103.72%
4B	16384	73.56%	+278.29%	+128.23%
4B	32768	78.93%	+374.52%	+158.34%

Each slash separates baseline and final means. The independent check verifies 12 requests, 221 receipts, and 101704 memory samples. Total controller time is 28.381 h; no request reaches its frozen time or memory limit. RSS samples include server load and the request at a nominal 1 s interval. The minimum sampled system availability is 11396.441 MiB; the maximum sampled server RSS is 6117.809 MiB. These observations apply to the tested nominal 16 GB board.

The capacity-extension report retains the complete protocol, sample ranges, archive identity, and repeat command. The previous report snapshot retains the pre-supplement report and its original claims.

6. Correctness and failure records

The final build passes 104 route, 95 attention, 432 SSM, and 158 GDN native cases. Its dump hashes match the verified RC1 reference. Two models produce 64 bitwise-equal full-vocabulary state records. These records cover RS 0/3, reset states, and prefixes 32/33/64/96 at context 4096. They do not replace long state acceptance.

Six complete natural answers cover original, RC1 C0, and final arms for both models. Each answer stops naturally and passes the frozen Unicode task's 106 functional checks. All 2B answers have identical text and 508 output tokens. The 4B original answer has 595 tokens and different text. The RC1 and final 4B answers have identical text and 521 tokens. Both 4B texts pass the functional checks.

The r1 delivery passes numerical checks but fails an argument-parser GPU-default assertion. The r2 delivery fixes that assertion but starts with four automatic slots. The r3 source fixes the server default to one slot and receives a fresh complete build. Both failed versions retain their source, build identities, and logs. No manual performance flag bypasses these checks.

The previous full-combination regression remains 3.0220477% for 4B/2048/65 against B1 under its original conditions. The new search does not erase that failed gate. The IME scalar schedule remains excluded after its failed performance screen. The final search tests eight combinations of B1-r2, C1, and C2-r2; none of them enables MTP. The earlier MTP pilot passes the relative quality gate on both models. Its six-task total time decreases by 8.63% on 2B and by 15.22% on 4B. MTP stays off because that pilot also applies an absolute usefulness gate. Direct decode fails the same gate on the interval-code task and on 2B arithmetic. The next plan tests MTP again on r3 with the relative quality gate and a per-category speed gate. GPU offload and page-allocation memory benefits remain unqualified. Predictor errors and wide intervals remain separate from measured model results.

7. Deployment and evidence package

The committee directory `final-release` collects runnable code, a one-command deployment entry, official conversion scripts, deployment guides, performance reports, and a design document. It includes complete production source and binaries, a fixed clean converter tree, model identity metadata, and separate evidence archives. The same production binaries retain the original verified hashes. The new entry creates tmux and acquires the fixed common lock before the server starts. Both model launches pass identity, actual-default, lock, and health checks without an inference request. The conversion scripts are included; the Scope section lists the full official conversion as an open item.

```
bash deploy.sh /absolute/path/to/model.gguf
```

The committee manifest, checklist, and file hashes describe its complete contents and retained acceptance limits. The submission plan maps all six requested materials.

The package contains the complete source, production binaries, fixed launcher, policy, dependency guide, and complete evidence. Both models use the same binary and defaults. Only the model path changes. Run this command in tmux from the extracted version directory:

```
bash package/with-lock.sh python3 package/start.py --model /path/to/model.gguf
```

Use `--spert` and `--tcm` only when dependency locations differ. These options select library locations, not performance profiles. The launcher checks binary, model, and runtime hashes. The final build passes two model launches after binary relocation.

The archive is `lab/final-default-20261013/release/k1-final-20261013-r3.tar.gz`. Its external sidecar records the archive SHA-256 without a self-reference. The evidence appendix records the final archive identity after the archive check. The Git text view omits large dumps, thread samples, and binaries. Strict receipt replay requires the retained complete archive. The local check verifies 338 comparison receipts and 119 final-validation receipts. The separate search check verifies 951 search receipts and 19 native receipts.

8. Historical evidence and remaining acceptance

The October 12 snapshot retains all previous report claims and caveats. Its evidence appendix and receipts preserve the previous record. Historical percentages with different controls must not be added or multiplied.

The October 7 release and October 13 RC1 remain comparison records. The new r3 package supplies the fixed default delivery for this campaign. The work ledger lists all work, the adopted items, and the reason for each stopped item. The next plan orders the remaining experiments by expected gain.

Scope

Configuration: MUSE-Pi-Pro, K1/X60, nominal 16 GB RAM, version `k1-final-20261013-r3`, 4 threads, batch 32, F16 K/V, one slot. Samples: 512 and 2048 inputs have three pairs per shape; 8192 and 32768 inputs have one pair per model; 16384 inputs have two pairs per model. The natural-answer check uses one Unicode code task per model and arm. The default is the balanced choice of the frozen search space. The overall comparison does not separate the contribution of each adopted path. The tested board has 16 GB of RAM.

- Pending: repeat stability at 8192, 16384, and 32768 inputs. Blocker: these shapes have one or two pairs. Release condition: measure at least three pairs per shape.
- Pending: general and multi-turn answer quality. Blocker: the fixed multi-turn quality suite has no run on r3. Release condition: run the suite on both models with the relative quality gate.
- Pending: long semantic quality and long state correctness. Blocker: the 16k and 32k requests use 65 forced output tokens. Release condition: run natural long-document answers and long state checks on r3.
- Pending: full official model origin and conversion. Blocker: the two official checkpoints have no complete download and conversion run. Release condition: convert both checkpoints with the included scripts, then compare files and answers with the current GGUF files.
- Pending: contribution of wide RVV, route 3, B1-r2, and C1. Blocker: the final comparison measures the whole version. Release condition: disable one path at a time on the same r3 binary and measure pairs.
- Pending: a unified public download archive for the exact dependencies. Blocker: the vendor runtime libraries are not in the package. Release condition: publish the exact runtime files with their SHA-256 values.
- Pending: operation on an 8 GB device. Blocker: no 8 GB board was tested. Release condition: run the capacity protocol on an 8 GB board.

Evidence appendix

Evidence cutoff: October 16, 2026, Asia/Singapore. This appendix supports the current report. The October 12 appendix retains the previous claim map in full.

1. Current claim map

Claim	Records	Check and limit
One compiled default for both models	Policy, build receipt	Source generator; 3271 source files
Actual default startup	Final validation, 2B log, 4B log	Default checker; no performance flags or experiment variables
Original compatibility boundary	Baseline receipt, compatibility patch	Original graph and compute kernels retain their bytes; MTP remains disabled
Frozen balanced selection	Search summary, mask decision, independent check	104 requests; mask 3 selected under the frozen 3% tie rule
Matched original-to-final performance	Comparison summary, independent summary	24 short requests and four long requests; each long shape has one pair
Natural answer quality	2B final, 4B final	106 checks per answer; one Unicode task does not establish general quality
Native and default state identity	Validation summary, state records	789 native cases and 64 short state records; long state remains pending
Preserved default failures	r1 check, r2 check	Parser assertion and automatic four-slot startup remain in the evidence
Previous report preserved	Snapshot receipts	Original Markdown, TeX, PDF, template, filter, and export script retain their bytes

2. Complete archive and compact text view

The archive path is `lab/final-default-20261013/release/k1-final-20261013-r3.tar.gz`. The archive SHA-256 is `d2adc88afcb3996047b6e925d40b0bcc0054a449f9616238f698eee1c092855a`. Its size is 1,503,683,264 bytes. The internal check verifies 5331 file receipts, 12 symbolic links, and 28 binary hashes. The archive receipt records the complete check result. The text evidence guide describes omitted large files. A receipt that names an omitted dump cannot be fully replayed from Git alone. The complete local evidence passes all recorded final checks.

3. Repeat the checks

Use the retained complete archive for strict receipt replay. Run checks in tmux. Board builds, inference, and tests must hold the fixed common lock. Do not modify old raw records.

```
python3 spacemit/bench/final-default/verify_final.py \
  --board lab/final-default-20261013/board \
  --delivery lab/final-default-20261013/delivery-r3 \
  --version r3 --output /tmp/k1-final-verification-new.json
```

The output file must not exist. The command above uses the complete local collection. An extracted archive stores production binaries only in its root `build/bin/`. Create the following internal link before a receipt replay from the extracted version directory:

```
ln -s ../../../build/bin evidence/final-r3/build/bin
python3 evidence/verification-tools/verify_final.py \
  --board evidence --delivery . --version r3 \
  --output /tmp/k1-extracted-verification-new.json
```

The link supplies the original evidence layout without a second production build. The checker verifies request order, identities, actual defaults, token history, cold counts, and sampled RSS. It preserves arithmetic means and observed ranges. The frozen selection replay uses `verify_search.py` and the original selected policy. The current package acceptance metadata differs from the pre-test manifest only in its acceptance status. The package receipt records that change. Source, policy, and binary bytes retain their tested hashes.

4. Acceptance and historical limits

Requirements.md controls acceptance. The gap map retains official model origin, conversion, long-state, and multi-turn gaps. The previous stop audit retains the 3.0220477% decode regression under its original conditions. The scalar schedule screen retains its failed performance gate. The TCM audit does not establish inference performance gains. Historical evidence remains dated and must not be combined into an unmatched speedup.

5. Final-build capacity supplement

The supplement report separates explicit capacity extension from ordinary default startup. The frozen protocol fixes actual 16384/32768 inputs and 65 outputs. The independent check verifies all 12 requests, 221 receipts, and 101704 memory samples. The controller summary records completion and no failures. The archive receipt records the separate supplement archive and file checks. Its complete layout supports an independent replay without the original local directory paths. The production archive and all previous raw records retain their original bytes. The pre-update snapshot receipt fixes the previous report set.

The supplement uses two pairs at 16384 inputs and one pair at 32768 inputs for each model. Only allocated capacity changes; optimization defaults and binaries retain their original identity. The ordinary context default remains 12288. Fixed-output completion does not establish long semantic quality, long state correctness, or formal multi-turn acceptance.

6. Committee deployment entry

The entry checks cover both fixed models. The receipts retain launch commands, effective defaults, properties, health replies, and common-lock records. The new entry sends no inference request. The first upload fails because scp rejects the directory name `.`; no model starts. Its log and exit status remain in the same evidence directory. A second upload uses explicit filenames and both model checks pass.

7. Default-build TCM investigation

The TCM report separates library access, hardware permissions, and buffer-copy routing. The independent check verifies two completed short requests and 13 process maps. Both server processes fail to open `/dev/tcm` with `EACCES`; neither maps that device. The source check matches four TCM-related files to the fixed vendor source. Ten local fixture tests pass for the user API. The board API rejects hardware allocation under the current permissions. Real hardware read/write and inference gains remain pending. The original production binaries and prior raw records retain their bytes.